



Product's behaviour recommendations using free text: an aspect based sentiment analysis approach

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Abstract

The assurance of astounding requirements and system specifications requires a tremendous exertion and requires semantically incorporated prerequisites. Particularly when utilizing diverse portrayals of requirements, the semantic integration is very troublesome. The present prerequisites management tools incorporate requirements just syntactically and, along these lines, give just constrained features to help the requirements analysis amid quality assurance. Sentiment analysis is a rising field of research and it assesses individuals' perspectives, thoughts or feelings. Amid the most recent decade, the research community concentrated on distinguishing and separating aspects. Some methodologies for aspect extraction depend on term frequency and frequency. These methodologies are very great if aspects are related to predefined classes and may fall flat if the concern is aspects with low frequency. However, the execution time for these systems increments because of the expansion in aspect set of data. Additionally, redundant and irrelevant aspects take an interest in deciding the assessment record related to the product, in this way precision of the algorithm is shifted. In this exploration, a segregational approach has been shown for perspective aspect extraction and aspect refinement utilizing similarity concept for the improvement of the quality of the product. To acquire preferable precision over existing methodologies we will additionally develop an arrangement of Visuword and integrate it with the proposed procedure. The experimental investigation will uncover that projected system beats the existing counter parts.

Keywords Quality assurance · Sentiment analysis · Aspect identification

1 Introduction

Reviews of the products have significant for forthcoming purchasers and help them to choose what to buy. At this point, where judging a review sentence's introduction i.e., negative or positive is considered as one of the prime

difficulties, distinctive opinion mining strategies have been proposed [1]. While these reviews are very useful in enhancing the quality of the product or overcome, the issues left in the product. These reviews may be useful to unravel the issues in the later version of that specific item. Due to rapid business changes and broadened mechanical advancement, numerous administrations are produced and devoured in a fleeting period [2]. In such a speedy and unusual condition, searching for manageability is essential. Thus, understanding what customers require is a fundamental task. As per the administration prevailing (S-D) rationale, the customer is constantly a co-producer of huge worth [3]. Since this esteem creation process is for the most parts drove in an online business focus in a continuous circumstance, how to appreciate customer necessities are believed to be fundamental. Particularly, consumers' wrangling authority is on a very basic level of extending, since consumers simply not diffuse information by online verbal trade [4], however furthermore share their

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impressions through a couple of kinds of online media, for instance, interpersonal organization administrations [5].

The fast growth of the World Wide Web (WWW) increases online communication. A huge number of customer's reviews or suggestions on everything are present on the web and these customers' reviews and suggestions are increasing day by day. Microblogging like Twitter, Facebook etc. is a powerful outlet for thoughts, suggestions, reviews and feelings of the people. Ranging from everyday observations to involved discussions, it is an enormous ever-growing source of texts. It is now estimated that, in just 60 s, over 400,000 Twitter posts are shared, about 300,000 Facebook statuses updated, about 25,000 items purchased from Amazon, over 5 million YouTube videos viewed and about 2.7 million Google searches are made among many other things. 153 million blogs are active and posted daily [6]. In addition to Epinions.com, rottentomatoes.com, and cnet.com which are opinion sites and their focus is a collection of amateur and professional reviews for several products and services. Before the rise of the internet to answer the question of "What do people think about any product or anything else", Surveys and polls are distributed in the form of paper in peoples. With the expeditious development of internet and the popularity of Micro-blogging sites like Facebook, Twitter etc. enables an alternative option for getting opinions from large population. Now a day's Web becomes the necessity for people to share their ideas, experiences and opinions as well as seeking others experiences and opinions [6]. Millions of ideas and experiences are shared every day, peoples can't read all ideas and experiences. About 2.7 million Google searches were made. A query "Artificial Intelligence" returns 98,400,000 results. This whole scenario demands fast, effective and accurate technique to track sentiments, opinions and ideas that are flowing on internet. Sentiment Classification is a key component of such techniques.

One of the most significant current issues in the user's research is providing an enhanced experience. To evaluate the usability and emotion, a process that improves user's experience should be required. People share their point of view in terms of sentiments. Sentiments are like and dislike, positive and negative, happy and sad etc. [7]. These reviews help different organizations to increase the quality of their specific product. Polarizing reviews in terms of sentiments is called sentiment classification. It is the process that automatically determines the sentiment category to which the textual content belongs [8]. We can categorize the reviews, comments and document in two types. These are numeric sentiment and categorical sentiment. Numeric sentiments include common example of numeric sentiment is a rating system in e-commerce sites. Using this rating system company judge the response of peoples. Categorical sentiment is the method to classify the comment or review

in different categories. These categories are binary (positive and negative), ternary (positive, negative and neutral) and multiple categories (Sad, Anger, happy etc.).

Opinions are become the necessary parts of all human activities as well as in the products to improve their quality. There are two types of reviewing sites, generic reviewing sites and specified reviewing sites. Generic reviewing sites include sites like amazon.com, epinions.com, rottentomatoes.com etc. and Specified-reviewing sites includes tripadvisor.com, yelp.com. Both of these reviewing sites have a significant effect in our decision meaning process. These decisions include buying a camera, smartphone etc., and making investments on any product etc., choosing schools, a decision about any movie etc. With the help of these reviews, the quality of the product could be improved through the application of the impact of these reviews in the product's development process. Before the Internet, other sources such as friends, relatives etc. affect the human decision process. Figure 1a, b shows the positive and negative review of Apple iPhone 7 from Amazon respectively.

The meaning of sentiment itself is still very wide. Opinion mining mostly focuses on opinions, which communicate or involve positive or negative sentiments from reviews, comments etc. These reviews are valuable to the quality assurance persons for taking intelligent decisions about the improvement of the product furthermore they are helpful for merchants knowing their item certain negative and positive characteristics [9]. Let's take an example that if somebody wants to buy a Mobile, such as a Samsung mobile, and is yet not confident about it, he or she can look on mobile sites and read peoples reviews about the mobile, after reading these reviews his decision will be relying on people reviews. This whole process is also named as

★★★★★ Perfect service and perfect iPhone 7plus

By Rex Allen on July 5, 2017

Size: 128 GB | Color: Red | Verified Purchase

This was a flawless transaction. I love my phone, it has been two weeks and the phone as described and was brand new. I love my iPhone 7plus. I would suggest this company

★☆☆☆☆ Completely disappointed

By Renata on June 28, 2017

Verified Purchase

The charger connector from the phone was broken, you have to jiggle it so it will hit the same spot on the screen to actually work. Completely disappointed

Fig. 1 a Positive review. b Negative review

opinion mining. History calculation sentiment of document is the task of marketing team of any company. Humans have no trouble in reading the movie review, product review and any political comment and categorized it in positive, negative or neutral class. The process of reading and understanding the descriptors which are placed on the subject of a sentence is used by the humans but when there are millions of reviews then it's a time-consuming task to read all reviews and categorized it in positive, negative or neutral class.

Sentiment classification or opinion mining has been studied at three different levels of classification [10]. These include aspect level, document level and sentence level. Document-level classification is the classification of whole document in a positive or negative. Sentence-level sentiment classification techniques read document sentence by sentence and decide which sentence gives a neutral, negative or positive opinion for a service, product etc. Aspect level or entity level sentiment classification is the most modern technique, which classifies the reviews or comments based on aspects or entities. When analyzing larger pieces of text using sentence level and document level sometimes does not give true reflection sentiments. This is where Aspect-Based Sentiment Classification (ABSC) comes in. With ABSC, you can dive deeper and analyze the sentiment in a piece of text toward specific aspects.

1.1 Research contribution

This research has the following objectives.

1. Produce an effective model to identify the quality issues of the product.
2. After successful deployment, the proposed model will find issues in the product to produce better quality with the help of user reviews.
3. The proposed model will help in determining the reason of the problem and investigating the engagement and amount of effort required to resolve the issue.
4. To develop a model that defines the way of fixing which would be most effective and upon making changes, consider possible backfires.

The structure of the paper is organized as: Sect. 2 describes the related works, Sect. 3 provides the detail of the research methodology, experimental results and analysis are discussed in Sect. 4, Sect. 5 elaborates conclusion and future work.

2 Related works

The literature related to the sentiment analysis and quality assurance has been conferred in this chapter. The literature has been observed concerning quality assurance using aspect based sentiment analysis. There are diverse methodologies, which expect to guarantee the quality of the product like consistency and completeness. A portion of these methodologies characterizes particular requirements or even transforms the requirements documentation specifically into product improvement. Moreover, no approach incorporates several representations of requirements semantically to gauge the quality attributes keeping in mind the end goal. A few methodologies incorporate the representations only syntactically.

Gewohn et al. [11] present a model based on quality for a client arranged perception of quality control and value data in industry gathering. The quality circumstance of an assessment article will be measured and assessed through quality and investigation attributes dependent on a vector show with an uncommon spotlight on the cooperation among man and framework for dynamic quality control. The goals of the quality model are to build item and creation quality just as revamp expense and quality expenses to accomplish the most elevated conceivable degree of generation proficiency. The quality model in this way offers a commitment for changing to a quality-situated savvy production line.

Khanvilkar et al. [12] exhibited the investigation of suppositions to break down the regular language and to discover the feelings, express by the people. The thought behind estimation investigation is to decide extremity of printed feeling given by individual. Notion Analysis is valuable in item proposals. In light of the audits given by the client; the items can be prescribed to another client. Real item sites are utilizing assessment investigation to comprehend the prevalence and estimations. The investigation for the most part defined as two class order issue, positive and negative. Slant investigation utilizing ordinal grouping gives all the clearer thought regarding arrangement. The framework will give extremity utilizing AI calculations SVM and Random Forest. The accomplished extremity will be utilized proposed framework decides extremity of surveys given by clients, utilizing ordinal issues with the item. Yussupova et al. [13] proposed an applied way to deal with the examination into consumer loyalty dependent on a point by point investigation of shopper audits written in common dialects utilizing artificial intelligence (AI) strategies, for example, Aspect Sentiment Analysis, Text Mining, Machine Learning and Data Mining. Extraordinary Internet assets for amassing client surveys, for example, tophotels.ru yelp.com and

tripadvisor.com are utilized as information foundations. To assess the viability of the projected methodology, they have completed a trial on subjective and quantitative research of lodging customer fulfillment. Indeed “Huge information” applications were considered as a probability to assess nature of administrations. The acquired outcomes demonstrate the adequacy of the proposed way to deal with choice help in item quality administration and contend for utilizing it rather than old style techniques for subjective and quantitative examination into consumer loyalty.

As far as the quality of the product is concerned, this process is performed during the construction of the product and the product has been checked step by step. The checking of the product has been performed on the basis of specification defined. Some aspects of a product show its weakness and shoppers feel disappointed or lesser compared to different items in a huge extent. Extant methods for item weakness detection depend upon tedious poll, so the productivity and precision are not fulfilled. An opinion-aware analytical framework to detect product weaknesses through sentiment analysis in a powerful way has been proposed by them [14].

Garca-Daz et al. [15] depicts in detail about the procedures utilized in the classification of the sentiment can be isolated into two gatherings: lexicon based methodologies and machine learning methodologies. In spite of the fact that the precision of lexicon-based methodologies at times can't give a precision practically identical to approaches of machine learning, the fundamental preferred standpoint of lexicon based methodologies is that it isn't required to have a lot of information to train and effectively utilized in numerous areas. Then again, machine-learning approaches as a rule need high computational expenses and a delegate training set. Therefore, if the proper development of mobile applications for sentiment analysis, lexicon based methodologies is the objective, because of the constraints of cell phone handling and the vitality utilization related with such processing in cell phones.

For software defect prediction, the vast majority of the current work accept machine learning and measurable strategies, for example, association control mining [16], support vector machine (SVM) [17], neural systems [18] and decision tree [19]. Nevertheless, software defect specific challenges have not considered by these studies. To start with, dataset for training utilized in investigations absences of all around programming modules well-labeled data for various causes. For software testing, gathering labeled software data is a costly errand given time constraint and restricted assets. Gathering labeled data for new projects or ventures is likewise hard with constrained advancement past.

Siddiqui et al. [20] briefed that in online customer feedbacks, automated analysis of the sentiments is

presented which facilitates both individual consumers' comparison shopping and organizations' business strategy development. An Ontology Based Product Review Miner (OBPRM) general architecture system is proposed by him. Based on the selected target product, user selects a specific product for opinion mining and product category is also defined by this system. To retrieve the customer reviews for a particular product, the OBPRM system uses API provided by internet search engines and e-commerce sites or uses web reviews. The extraction of ontology is performed offline and before the conduction of context-sensitive mining. The taxonomic information like “iPhone” or product “is-a” mobile phone is captured by the fuzzy domain ontology which is about product category. It also captures the relationship like “Screen” which is associated with the product features and also known as non-taxonomic relationship. The context-sensitive mining system development, evaluation of the novel fuzzy domain, ontology and design would be illustrated through this.

Diao et al. [21] presented the identification of the feature as method utilized for the deduction of product features out of the texts that are tagged. Generally, for giving names to the real word entities which are nouns POS is utilized. Products and its features like image quality, zoom and battery life, a noun give names in this case. Frequent and infrequent features are defined as two categories of features in these works. In their study for opinion mining, an ontology based approach was proposed. In both computer-process and human-understandable way, this portrays the domains' semantic. In the area of information extraction (IE), this would be motivated by its success. A fine-grained approach which used ontology structure for opinion mining as vital part of the extraction of the features process by considering the relations between concepts is introduced in this paper. The benefit of introducing ontology to opinion mining is clearly shown in the experimental results by them.

Hu et al. [22] proposed an approach by combining methods from data mining and text mining which allows automatic analysis of customer experience from hotel reviews. There are four main phases of the system which are sequentially executed and named as selection, extraction then aggregation and finally product features analysis. For product development they gained valuable information and evaluations are also considered. The author chooses supervised learning with training examples for the recognition of product features and with explicit features in the current work. The product features along with their sentence valuations are connected by it and by depicting evaluations and aggregation of products it is summarized. By identifying the major determinants of the customers' satisfaction and using association rules by applying

decision trees with the product on the whole, the analysis component detects dependencies among product features.

Lamba et al. [23] explored that AI is a division of Artificial Intelligence which constructs a framework that gains from the information. AI has the ability of taking the crude information from the archive which can do the calculation and can foresee the product bug. It is constantly attractive to distinguish the product bug at the most punctual with the goal that time and cost can be decreased. Highlight choice system wrapper and channel technique are utilized to locate the most ideal programming measurements. The principle point of the paper is to locate the best model for the product bug expectation. In this paper AI systems like Linear Regression, Random Forest, Neural Network, Support Vector Machine, Decision Tree, and Decision Stump are utilized and near examination have been finished utilizing execution parameters.

Qiao et al. [24] demonstrated that exertion mindful in the nick of time imperfection forecast is to rank source code changes dependent on the probability of recognizes just as the push to review such changes. Exact defect prediction calculations help to discover more imperfections with restricted exertion. To improve the precision of imperfection forecast, in this paper, they propose a profound learning based methodology for exertion mindful without a moment to spare defect prediction. Assessment results on a notable informational index recommend that the proposed methodology in their paper beats the best in class approaches on every one of the subject activities. It improves the normal review and pop by 15.6% and 8.1%, separately.

From the above discussion it is concluded that user assessment can be considered as the valuable asset for enhancing the nature of the item yet shockingly there is absence of machine learning methods, which use the reviews to enhance the quality assurance process for better products [25–27]. Moreover, the research problems cited above may require more refined quality assurance technique. The proposed technique will help to improve the quality of the product through user reviews.

3 Research methodology

For the achievement of the progressively effective procedure, make sure that the proposed methodology should beat the existing counterparts in all aspects for improving quality assurance process using sentiment analysis. In the process which is famously known as sentiment analysis especially in aspect extraction, the proposed technique has been compared with the existing approaches which have better response and credibility. The proposed conceptual framework has been discussed in this section comprises of

aspect and opinion extraction which further leads to aspect refinement and aspect score as shown in Fig. 2. The tabular and graphical representation of each and every phase has been provided to help out the reader for enhanced understanding of the complete process of the research.

3.1 Dataset description

What dataset is used and how it is prepared? This is the common question which needs to be answered. To answer this question, following sub sections have enough information. Based on terms and conditions with the end goal of usage and protection safeguarding, reviews of cars and hotels have been collected from Tripadvisor based on 259,000 reviews and Edmunds consist of reviews which are 42,230 in number. Full textual review along with the favorites, dates and author names are in the extracted fields for cars. Gold standard judgment for ranking has been included for hotels in addition to the fields like title of the review and date. The third dataset is composed of the information about the Mobile phone named as Samsung Galaxy S8+ included 3700 user reviews in it. The detail description of datasets is as under.

3.1.1 Car reviews dataset

The car review dataset consists of 2015, 2016, and 2017 model car's full reviews. Each model year has around 140–250 cars. Full textual review along with the favorites, dates and author names are the part of extracted fields. Table 1 shows the detailed description of car dataset.

3.1.2 Hotels review dataset description

There are reviews of around ten hotels from all over the world choose from the ten famous cities i.e., Beijing, Dubai, New York City, London, San Francisco, New Delhi, Las Vegas, Shanghai, Chicago, Montreal, Abu Dhabi and Riyadh. In each city there are about 90–650 hotels. Date, review title and the full review are included in the extracted fields. There are total 259,000 reviews

3.1.3 Mobile phone review dataset description

The mobile phone dataset is about the famous mobile phone names as Samsung Galaxy S8 + and consist of 3700 number of reviews. Full reviews of mobile phone's functionality like screen size, battery, android version have been available in the dataset. Vocabulary size of around 1526 and the words in lexicon are 932.

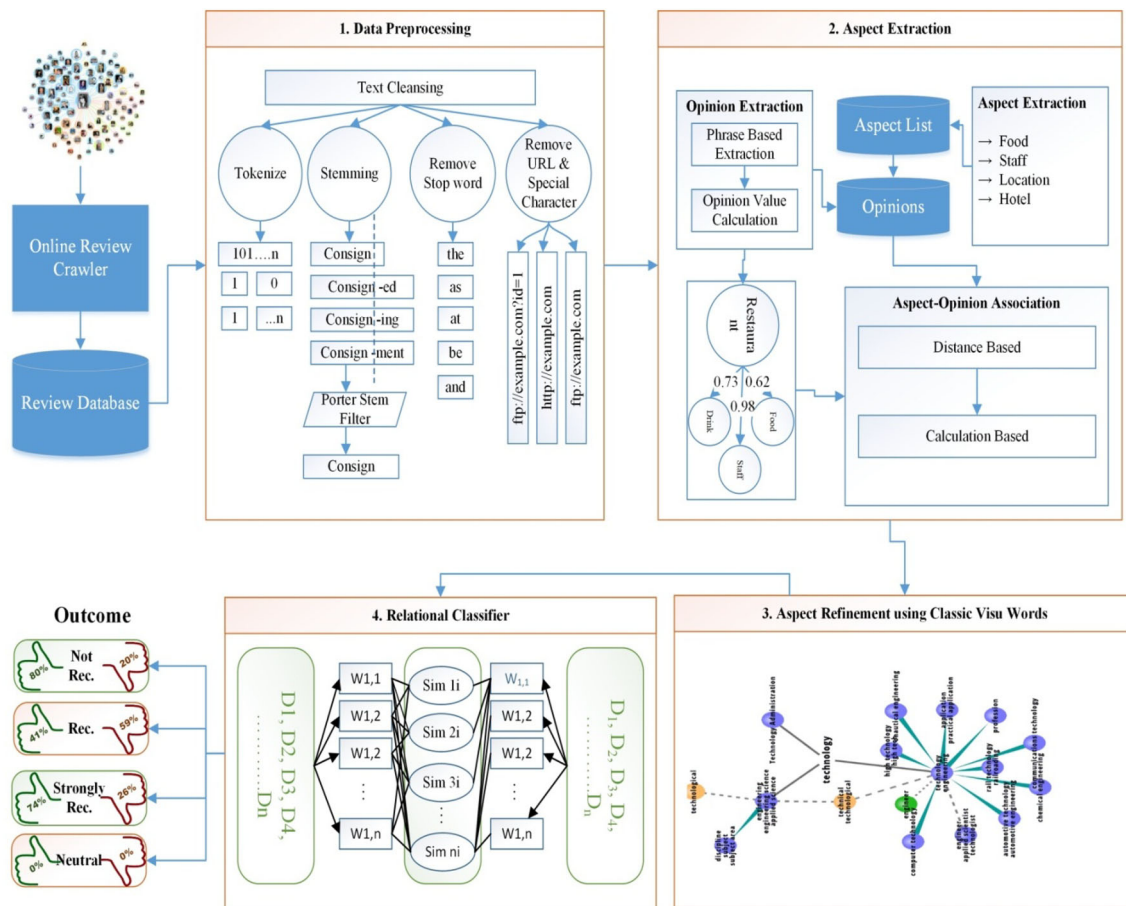


Fig. 2 Proposed conceptual framework

Table 1 Car reviews dataset

Dataset name	Year	Number of reviews	Vocabulary size
Car reviews dataset	2015	18,903	8200
	2016	15,438	6100
	2017	7947	2725
	Total	42,230	17,025

3.2 Preprocessing

The phase of preprocessing is expressed in Algorithm 1. The detail of the whole phase is clearly mentioned in the algorithm that how the input is added in the form of text and then after tokenization, lowercasing, normalization, stemming, removal of irrelevant contents and transliteration process the output is obtained, e.g., “the car is not comfortable car” takes the word with the extremity of positive but the unacceptable appearance i.e., “NOT” is the reason of changing the meaning of the sentence into negative.

Algorithm 1: Preprocessing

```

1. Input: A Set of Reviews ( $D_i$ )
2. Output: All Preprocessed Words ( $W^{all}$ )
3. Initialize  $W^{all} \leftarrow \emptyset$ 
4. for all  $D \in D_i$ 
5.     String  $S$ 
6.      $S \leftarrow \text{tokenization}(D_i)$ 
7.      $S' \leftarrow \text{lowercasing}(S)$ 
8.     end for
9.     for all  $S_i \in S$ 
10.         $S'' \leftarrow \text{Normalization}(S_i)$ 
11.         $S''' \leftarrow \text{Stemming}(S'')$ 
12.         $S'''' \leftarrow \text{Stemming}(S''')$ 
13.         $S''''' \leftarrow \text{Transliteration}(S''''')$ 
14.    end for
15.    return  $W^{all}$ 
end

```

3.3 Aspect and opinion extraction

In this section, a procedure for aspect and opinion extraction dependent on aspect and assessment words unraveling and affiliated likeness alongside grammatical feature tagger has proposed and connected to each word in an obstinate sentence to think of it as either as a viewpoint or non-viewpoint word. By utilizing aspect and opinion extraction technique, the proposed philosophy overrides the current best in class approaches in accomplishing high exactness of practically 99%. The proposed model for aspect and opinion extraction is shown in Fig. 3 which clearly demonstrates the process of aspect and opinion extraction along with the utilization of opinion lexicon for the calculation of opinion value.

In the first step of the model, the aspects were collected in base list and then the model was produced based on the aspect and entity model for the organization of connected aspects and entities. In an automated way this model was generated. Later, the extension of the list has been performed and then in the last step, by lemmatizing all words that list was normalized. Table 2 shows some entities and aspect groups as well as examples for synonyms and sub-aspects.

Opinion extraction is the second assignment of the extraction procedure. The extracting of the corresponding opinion words used for every product aspect by the customer is involved in this task. What customers think about product aspects are the opinion words and they are in the form of adjectives. Opinion words are usually placed in the sentence near to aspects. Opinion words are placed as the

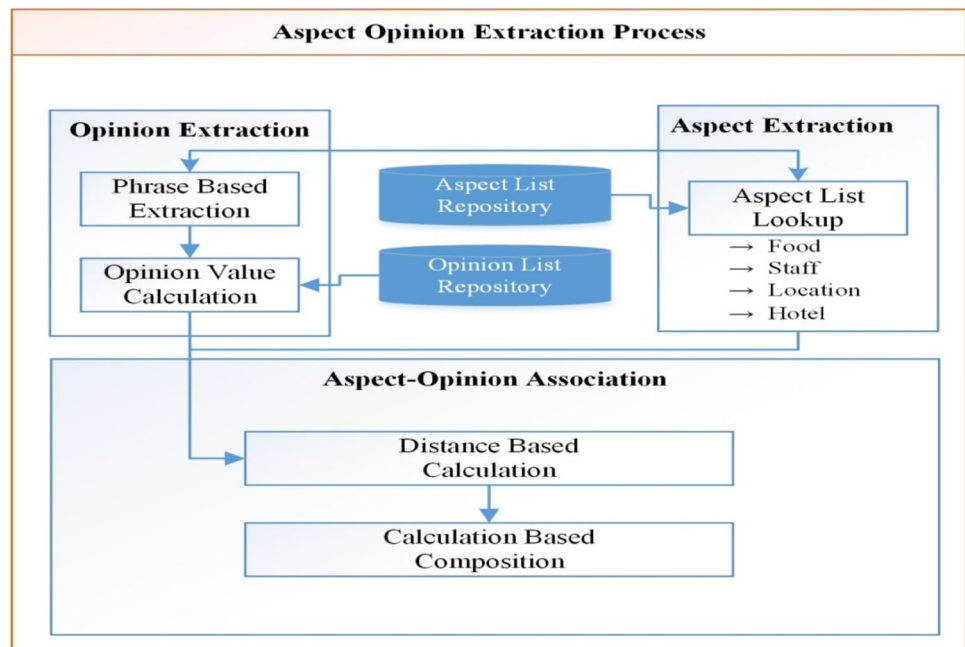
Table 2 Some aspect groups in car and hotel domain

Aspect groups	Examples for sub-aspects
Product	Automobile, hotel
Costs	Car price, room rate, discount
Service	Service staff, customer service
Conditions	Contract, offers, warranty

closest adjective to the aspects by some researchers [28]. The lemmatizing of the phrases is essential to get the opinion value from the lexicon due to the fact, that in the opinion lexicon all words are lemmatized. To lemmatize all words in the right way there are some minor changes which need to be applied to the procedure and specially the lemmatizing for comparative and superlative forms to suppress. Nevertheless, first the opinions words have been located in the sentence and from there the corresponding aspects have been determined by searching the sentence backward first for the closest aspect, if did not find, then search goes forward.

The opinion lexicon incorporates opinion values for both, single supposition bearing words and expressions with a length of up to five words. While utilizing the sentiment dictionary, in the majority of the cases it is conceivable to acquire supposition esteem for a given assessment bearing expression separated from the content straightforwardly. On the off chance that the expression comprises of more than a single word, the expression is abbreviated by single word and another query in the supposition dictionary is performed. Towards the end, every

Fig. 3 Aspect and opinion extraction process



phrase which contained opinion is sorted as strong positive, if the conclusion worth is more noteworthy than 0.70 and as strong negative, if the sentiment worth found is littler than -0.70 . Now, values of the opinion can be ordered into three equivalent estimated interims (negative, impartial and positive) and that both, the positive and negative class, can be isolated again into subclasses speaking polarities of weak and strong.

Table 3 lists examples of strong positive and strong negative opinion bearing phrases with their opinion values. Adjective- and noun-based phrases have been distinguished. Also shown is one example of an opinion phrase which has an opinion value of less than 0.66, thus being not regarded as strong positive.

Using a distance-based approach applied on the sentence-level, the linking of the opinion phrases to the aspects is done. To the next aspect found in a sentence according to the word position, all strong positive or negative opinion phrases are linked. If opinion phrases are more than aspects like two opinion phrases and one aspect, then link to the aspect is only the nearest phrase.

3.4 Aspect reduction

Aspects are purified through the Visuwords™. Promising results are observed clearly from the evaluation and the rise of 5% to 6% has been observed in comparison with other techniques, increasing the aspect refinement process optimality. Visuwords™ is an online graphical dictionary used for look up words to find their meanings and associations with other words and concepts. It makes nodes with all its associated words and the meaning and each detail of the word is available on the node. Touch a node to see the definition of that word group and click and drag individual

Table 3 Words and phrases with their value of opinion and categorization

Adjective-based phrases	OV	sp/sn
Great	0.93	sp
Very little valued	0.88	sp
Knowledgeable	0.75	sp
Caring	0.57	–
Worthless	-0.70	sn
Just wicked	-0.85	sn
Noun-based Phrases	OV	sn/sp
Emotion	0.85	sp
Bluff bundle	-0.73	sn
Impoliteness	-0.94	sn

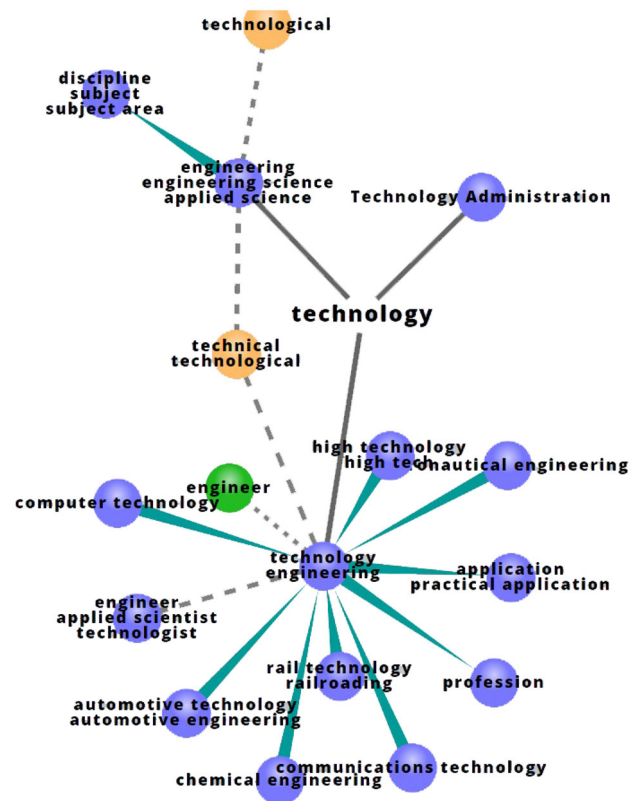


Fig. 4 Working of Visuwords™ for aspect refinement

nodes to move them around to help clarify connections. Figure 4 represents the working of Visuwords™.

A network of nodes or ‘synsets’ will spring out from the word that you entered in the search bar and create the network of the nodes. By several terms or synonyms, a synset is a single concept that is represented. The words which convey the same idea but different in spelling is known as synonyms. For instance, when you type the word “see”, it has been noticed that there are number of synsets appeared for the word and one more thing that each synset is represented by a green circle. So all synsets in green represents the verbs. Each of the synse has its definition and when the node is touched the definition of the word has been displayed.

The sysnet act as a globe in which blue is for nouns, green is for verbs, red is for adverbs and orange is for adjectives. Through the links of the color the synsets are joined which shows the kind of an association with each other.

- “is a kind of”—hyponym/hypernym pairwith regards to “wheat” and “grain”, we see a cyan link from “wheat” pointing towards “grain” we can understand this to mean that wheat “is a kind of” grain. Here, “wheat” is a hyponym and “grain” is a hypernym.
- “is an instance of”—pair of hypernym/hyponym

- “is a substance of” “is a member of” and “is a part of”—pair of holonym/meronym pair

After this step refined aspect has been obtained through which we are able to find the issues in the product related to the quality improvement and could be used in product weakness classification.

3.5 Product improvement recommendation

When the final results have been obtained, there is need of a recommendation for the process improvement. The relational classifier has been used at this place which evaluates the probabilities of the class exclusively based on the same type of the entities whose labels of the class are recognized. There are two strong assumptions based on which this classifier works. The first one is the labels of class which are known within the same structure that is linked with some entities. The second is the homophily exhibition of the entities which are related to each other and through one or more dimension related to the same class [29]. The performance of the classifier may not be good in case when there are isolated entities or labels are not recognized.

The relational-neighbor classifier estimates $P(c|e)$, the class-membership probability of an entity e belonging to class c , as the (weighted) proportion of entities in De that belong to class c . We define De as the set of entities that are linked to e . Thus,

$$P(c|e) = \frac{1}{Z} \sum_{\{ef \in De | \text{label}(ef)=c\}} w(e, ej) \quad (1)$$

where $Z = \sum w(e, ei)$ and $w(e, ei)$ is the weight of the link between entities e and ei . Entities in De that are not of the same type as e are ignored. If De is empty or has no entities with known class labels, then the RN will estimate e based on the class prior (of the known labels).

At the end, the suggestion would be given on the basis of three predefined classes named as Not Recommended, Recommended and Strongly Recommended. Furthermore, Not Recommended class would suggest that it is not recommended to do these specified changes for the improvement of product quality based on user review while the Recommended class would show that it is recommended to do these changes for the improvement of the product. Last but not the least, the Strongly Recommended class would suggest the changes that need to be improved or implement on priority basis and without these the quality of the product is not improved according to the standards on the basis of which that specific product has been created. Finally, there will be a neutral class in which there would be results with neutral review.

4 Experimental results and analysis

In this section, for the evaluation of the proposed conceptual framework, on different datasets various experiments have been performed. In terms of precision, recall and F-measure, the first experiment is for the evaluation of aspect and opinion extraction process. The second experiment consists of the comparison of VisuwordsTM with other techniques for the aspect refinement using word association while the last experiment portrays the novelty of using the relational classifier for the recommendation to improve the quality assurance process in terms of accuracy by comparing with existing approaches.

4.1 Experimental evaluation of aspect extraction

By the proposed technique, promising results have been achieved. Cars and Hotel dataset have been used for the experiments. The results of comparison between proposed approach and traditional POS are shown in Tables 4 and 5. Precision, recall and F-measure have been adopted to evaluate the efficiency of proposed approach. The number of successfully evaluated aspects set for a given review document divided by the number of all aspects is known as the precision like only 20% of the aspects are correct in case where 100 aspects have been assessed. $20/100 = 0.8$ is defined as the precision. The number of successful evaluated aspects for a given review document divided by the total number of aspects is known as recall like in the case only 18 of the aspects are correctly evaluated where 1000 aspects have been reserved so $18/100 = 0.18$. The harmonic mean of the precision and recall or it is roughly the average for close values and it incline to the lower for distinct values known as F-measure.

With the standard POS and then with proposed approach when applied on car dataset. Table 4 clearly indicates in terms of precision, recall and F-measure the performance of proposed approach. Initially, the recorded precision is 82.74% when standard POS is applied while 75.15% is the recall reached and the output of 78.76% is recorded for F-measure. It is obviously seen that the estimations of Recall, Precision and F-measure are more encouraging than the past one.

Table 4 Comparison of proposed approach with standard POS for car dataset

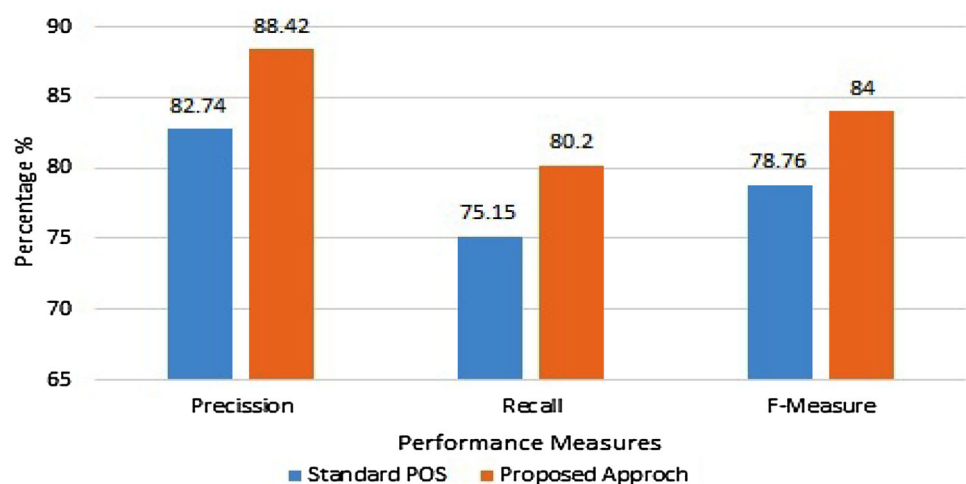
Measures	Proposed approach	Standard POS
Precision	88.42	82.74
Recall	80.2	75.15
F-measure	84.0	78.76

Table 5 Comparison of proposed approach with standard POS for hotel dataset

Measures	Proposed approach	Standard POS
Precision	80.65	85.42
Recall	85.37	77.21
F-measure	83.0	81.10

In the table for Hotel dataset i.e. Table 5, to evaluate the performance of the proposed approach again the precision, recall and F-measure have been calculated. The table clearly depicts that on the Hotel dataset when standard POS has applied it gains the value of 85.42% in precision, 77.21% in recall and 81.10% in F-measure. 80.65% is with precision, 85.37% with recall and 83% with F-score have been achieved when proposed approach is applied on the given dataset showing the overall performance of the proposed approach. The graphical representation is shown in Fig. 5.

Last but not least, for mobile phone dataset when precision, recall and F-measure has been calculated to evaluate the performance of the proposed approach. The graphical representation in Fig. 4 clearly demonstrates that the Standard POS provides for precision the output of 86.01%, for recall 81.32 and for F-measure 83.59. While the proposed approach performed well with output of 91.15% for precision, 88.47% for recall and 89% for F-measure which is quite remarkable. In figure there is a representation of both standard POS and Proposed Approach (PA) along with the representation of precision, recall and F-measure in the form of bars which clearly demonstrates the results in percentage. The graphical representation depicts that the proposed approach beats the standard POS in all fields when it comes to check the precision, recall and F-measure.

Fig. 5 Comparison of proposed approach and POS in terms of precision, recall and F-measure

4.2 Experimental evaluation of aspect reduction

The second experiment consists of the comparison of VisuwordsTM with other state-of-the-art techniques for aspect refinement using word association. In this process, most relevant aspects have been kept and the least relevant would be terminated with the help of VisuwordsTM. It has been clearly observed that the proposed method for the aspect refinement is up to the mark and gives promising results when compared with other techniques of aspect refinement. The comparison of the proposed technique has competed with the hybrid approach [30]. In this approach, a hybrid technique for aspect identification has been designed. Comparison of proposed aspect extraction technique with a hybrid approach is shown in Table 6.

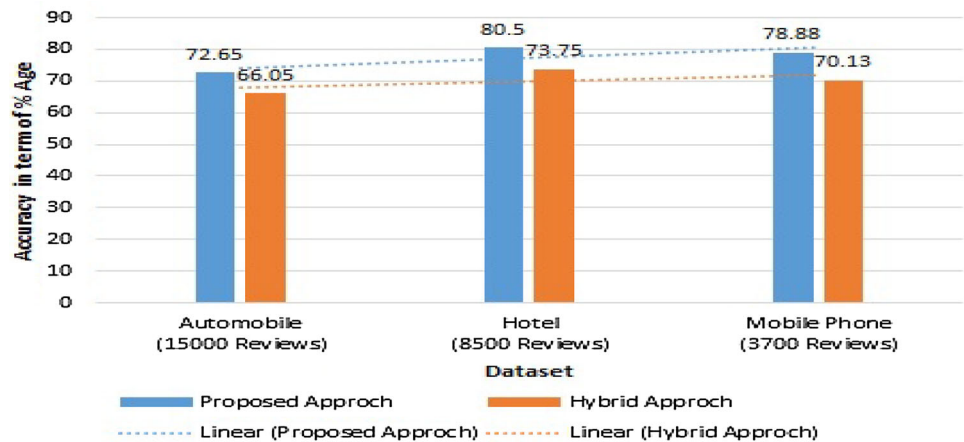
It is clearly shown from the above table that proposed technique improved the accuracy up to 6.5% in Car dataset, 7.3% in Hotel dataset and 6.1% in Mobile phone dataset. It has been demonstrated that the accuracy of the proposed technique on Cars and Hotel dataset is more than 70%, and 79%, respectively on a standard benchmark dataset. The graphical representation is shown in Fig. 6. The main reason is that the hybrid approach used heuristic combination in part-of-speech (POS) patterns for detecting explicit single and multi-word aspects. Therefore, the proposed approach performs well in this scenario.

4.3 Experimental evaluation of product improvement recommendation

In last experiment, the comparison of the proposed approach using relational classifier has been performed with the multilevel relational classification (MRC) [31]. It has been observed that the proposed approach with relational classifier performs up to the mark. The datasets used in this comparison are diverse and cover different type of

Table 6 Proposed approach comparison with hybrid approach

Dataset	No. of reviews	Proposed approach (PA)	Hybrid approach (HA)
Car	15,000	72.65	66.05
Hotel	8500	80.50	73.75
Mobile phone	3700	78.88	70.13

Fig. 6 Comparison of proposed and hybrid approach

cars along with their features and specifications plus the hotel data set which has also been utilized for comparison as shown in Table 7. It has been evaluated from the accuracy comparison of the proposed technique, it's quite amazing to observe that on Car dataset the relational classifier gains the higher accuracy of 87.45% as compared to the MRC which is 81.84% yielding almost 6.4% higher accuracy. The raise has been observed due to the property of relational classifier based on correlation based on word association used in proposed approach.

In the comparison of MRC and proposed technique on Hotel dataset, it is vigorously demonstrated that the proposed technique obtained 91.56% accuracy in comparison with MRC which is 86.24%. Last but not least, the Mobile Phone dataset, again the proposed approach is leading with 90.85% while the MRC scored 85.78%. The graphical representation is shown in Fig. 7. Since the MRC works uses POS patterns for detecting single and multiword aspects, therefore some of the aspects are not covered and thus may lead to lower accuracy results in particular datasets.

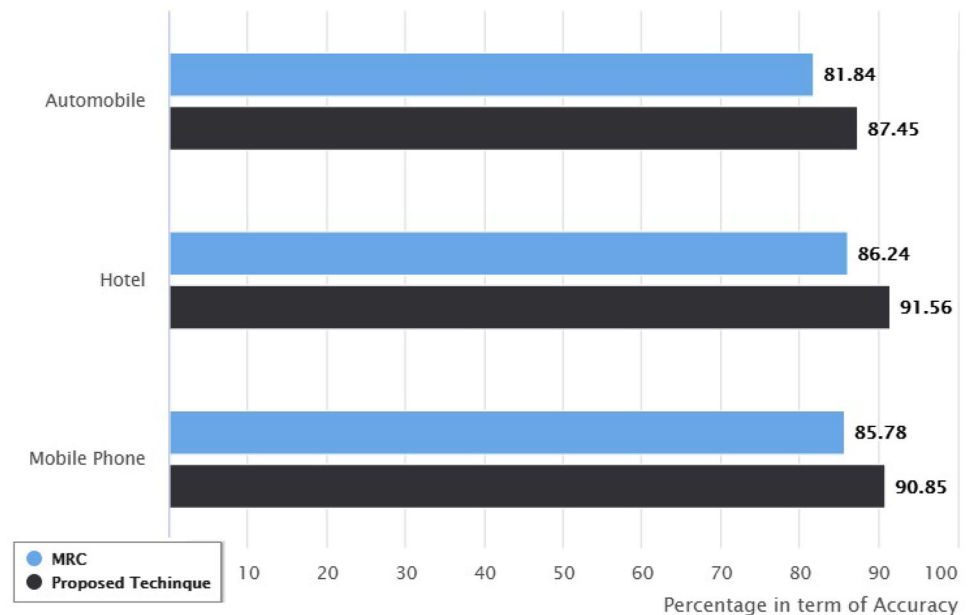
Table 7 MRC and P.T comparison results

Dataset	MRC	Proposed technique (P.T)
Cars	81.84	87.45
Hotel	86.24	91.56
Mobile phone	85.78	90.85

5 Conclusion and future work

Sentiments Analysis helps in, improvement of the quality assurance process which raises the quality of the product is an emerging pitch of assessment which estimates the product through people's opinions, thoughts or feelings. In classification task due to the huge amount of content which is issues and generated on the internet, it plays a key role. From the number of application, its performance is flawless and depicting the benefits of this research. To improve the quality of the product, this research proposes a novel technique. The proposed conceptual framework has been discussed in this chapter comprises of aspect and opinion extraction which further leads to aspect refinement and aspect score. The tabular and graphical representation of each and every phase has been provided to help out the reader for enhanced understanding of the complete process of the research. To evaluate the performance of proposed approach, various experiments have been performed. Two webometric measures have been adopted in the first experiment. To refine the aspect extraction process, each of the chosen method is used. Different views of evaluation of aspects represent the output of each measure. The proposed method gives are markable output when it comes to the comparison with the different existing approaches is clearly shown in this case. It works well in all respective areas as observed. In comparison with MRC hybrid approach, the obtained results are quite promising for the proposed approach. In future, the proposed model could be enhanced for real time scenarios.

Fig. 7 Comparison of MRC and proposed technique



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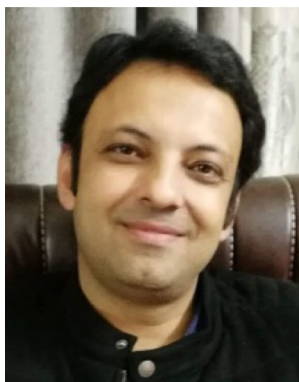
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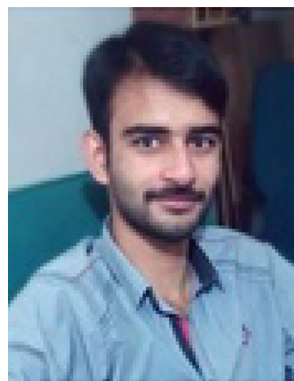
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